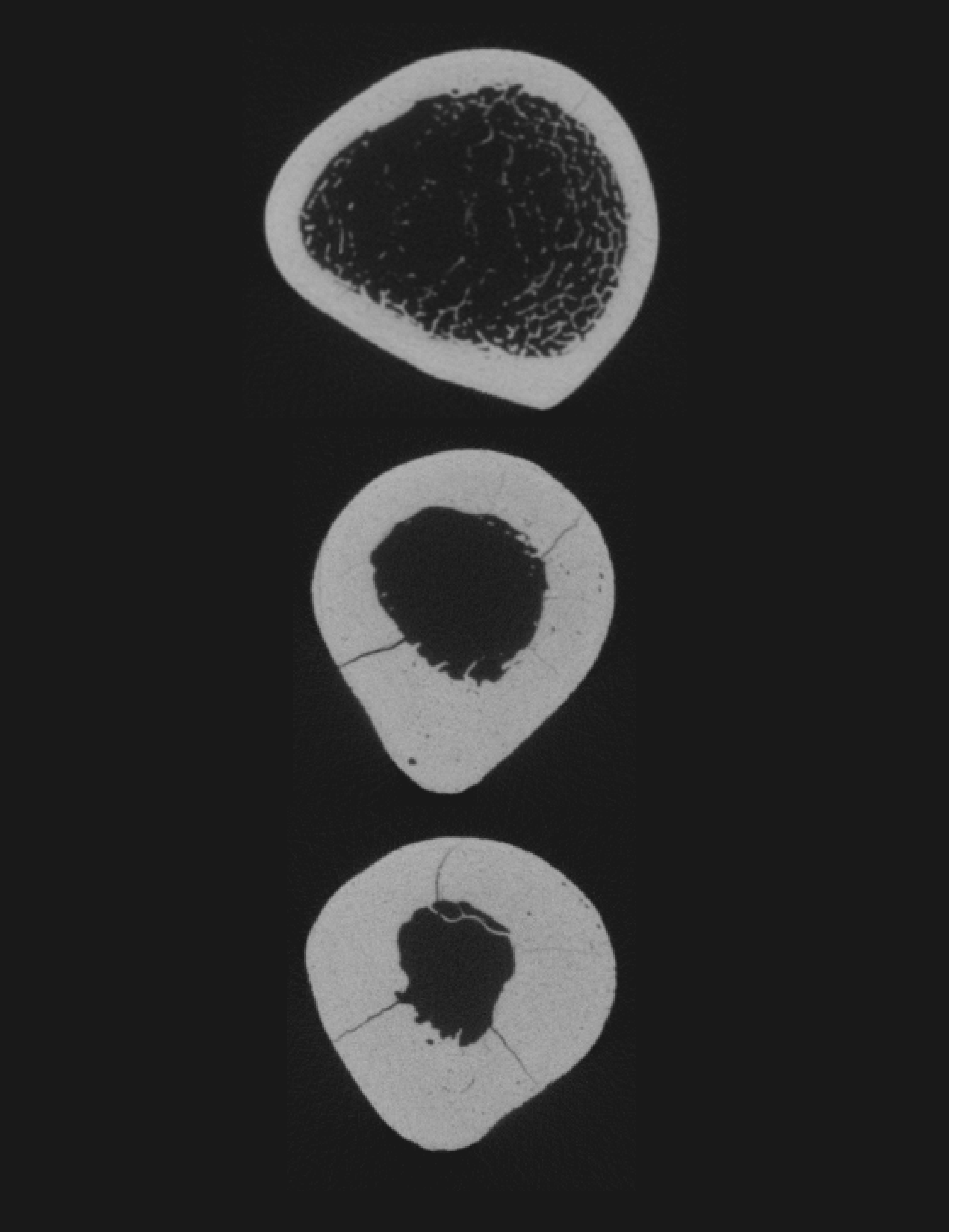
Skeletal Response to Activity



Objectives

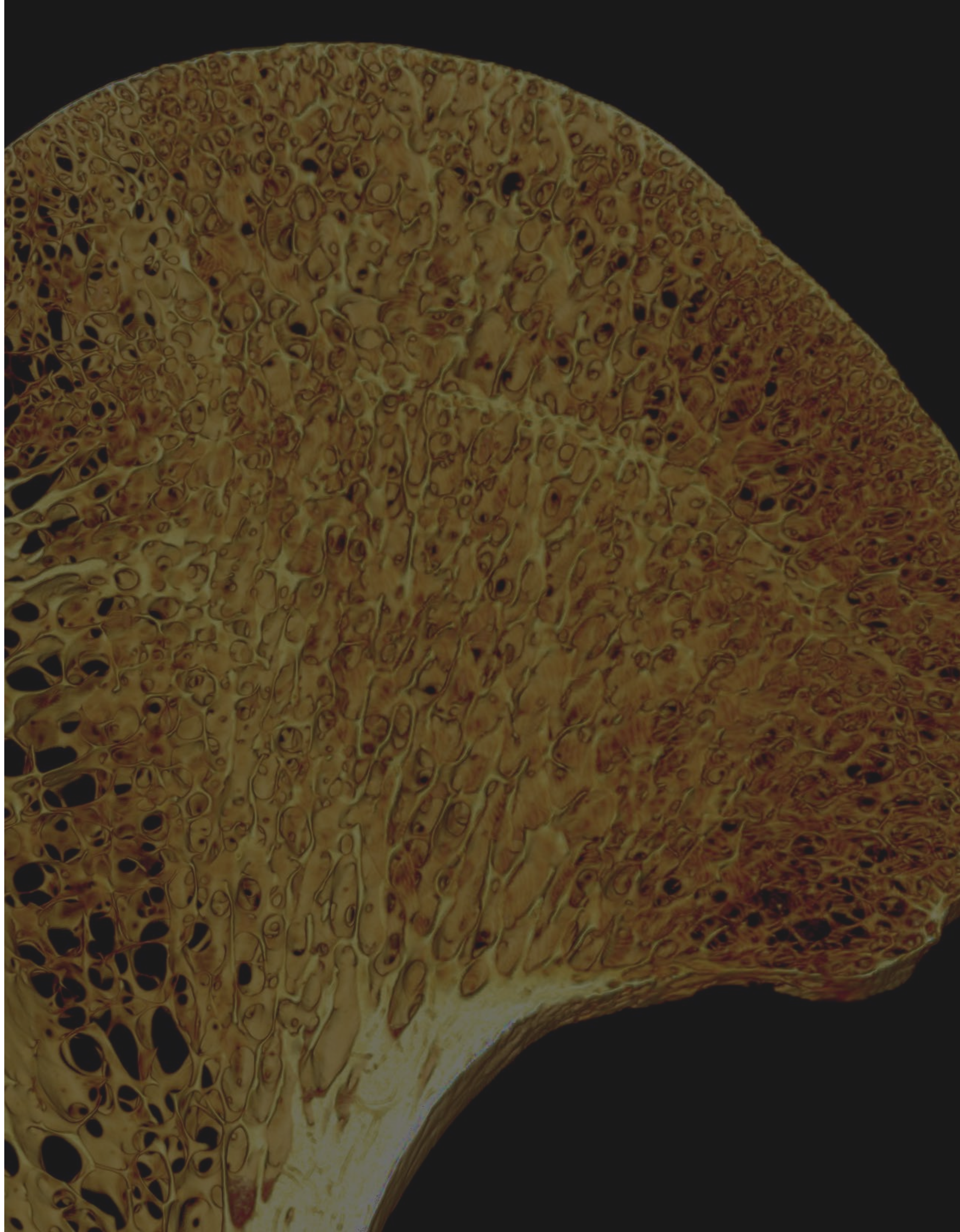
1. Recognize how the skeleton responds to activity and loading

2. Describe how engineering concepts can be used to interpret skeletal structures

3. Apply engineering concepts to analyze skeletal variation and its relationship to animal behavior

Outline

- Skeletal plasticity
- Bone functional adaptation
- How are bones loaded?
- Engineering beam theory
- How do we measure bone shape?
- Does bone shape reflect loads from behavior?

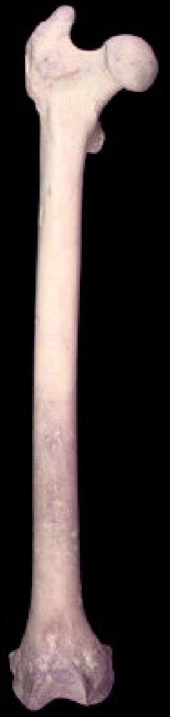


Adaptation at different scales

- Many morphological features are driven by selection
  - Whole bone morphology
  - Limb length
  - Olecranon process length

- Other morphologies are plastic and respond to behavior/activity



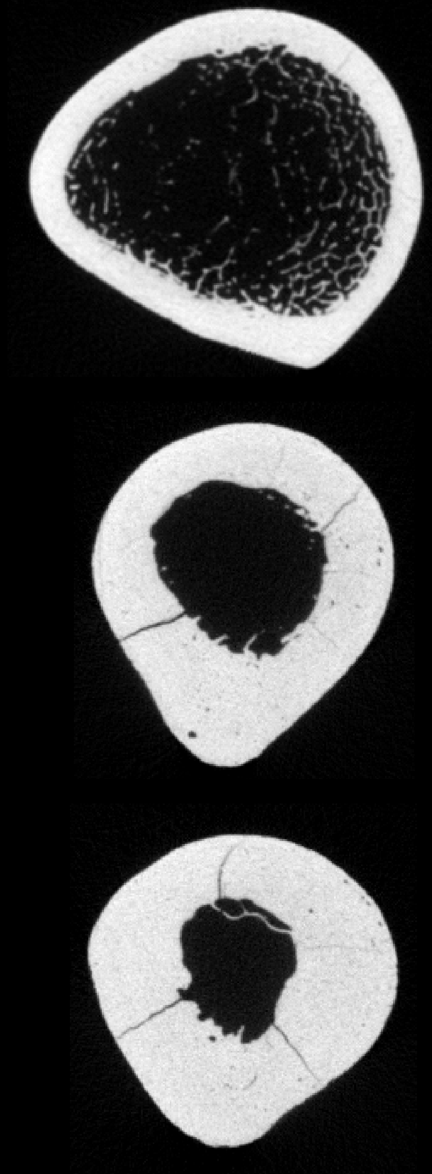


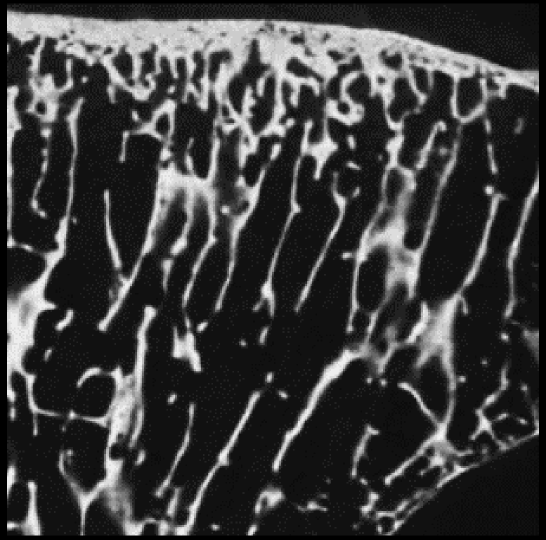

Bone is a living tissue

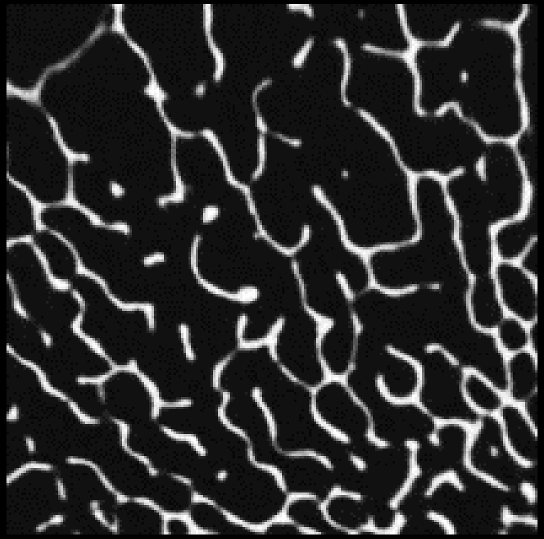
- Skeleton responds to environmental stimuli
    - Diet and nutrition
    - Activity
    - Pathology and disease
    - Climate

- Biological concept: **plasticity**

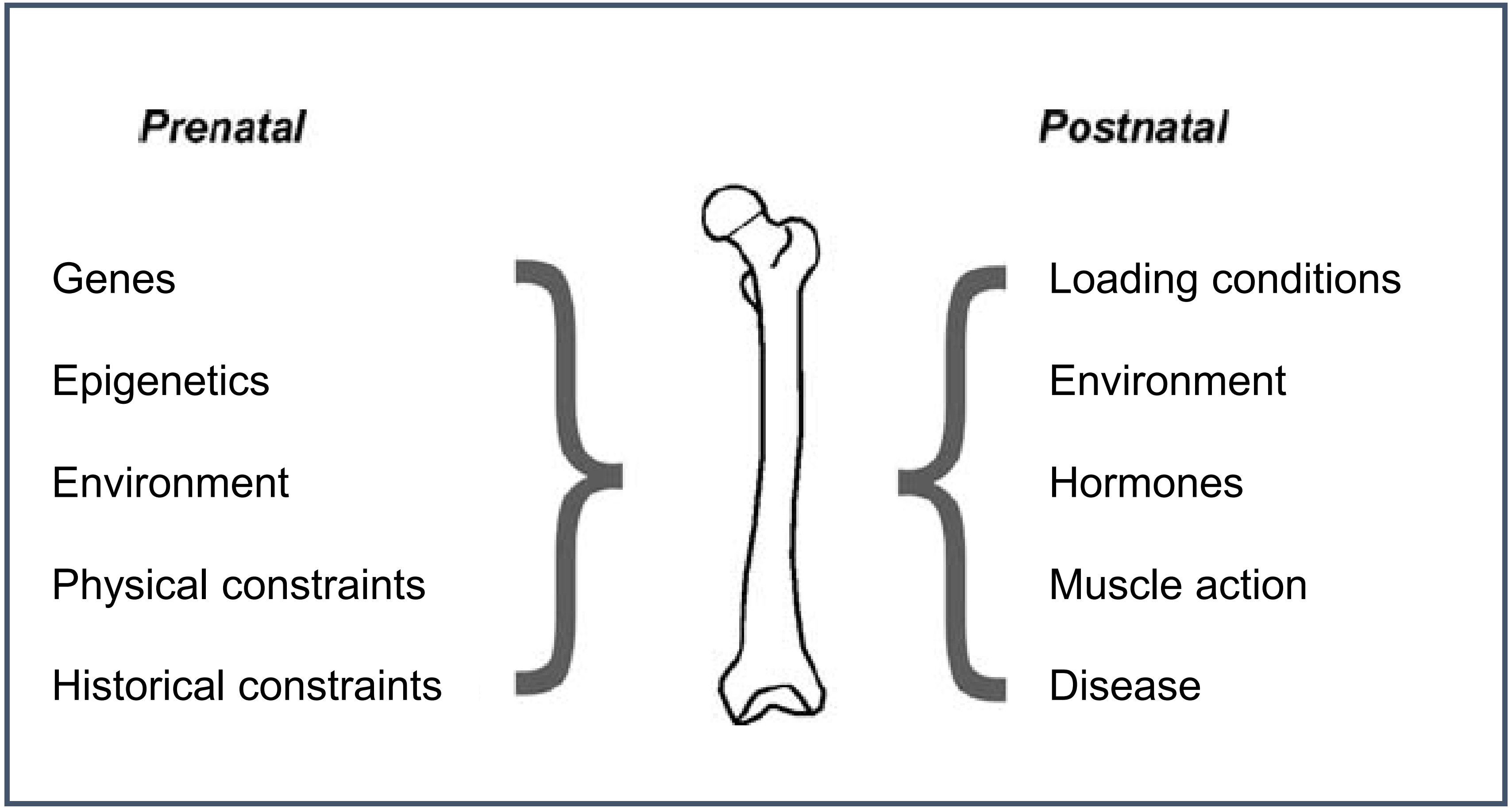
- Biomechanics of bone are important for anthropology







Many factors influence skeletal morphology



Macho 2007

Bone Functional Adaptation

- Wolff's Law
- Link between bone structure and the magnitude and orientation of loads on the skeleton
- Mechanisms are complicated

Why is this phenomenon important?

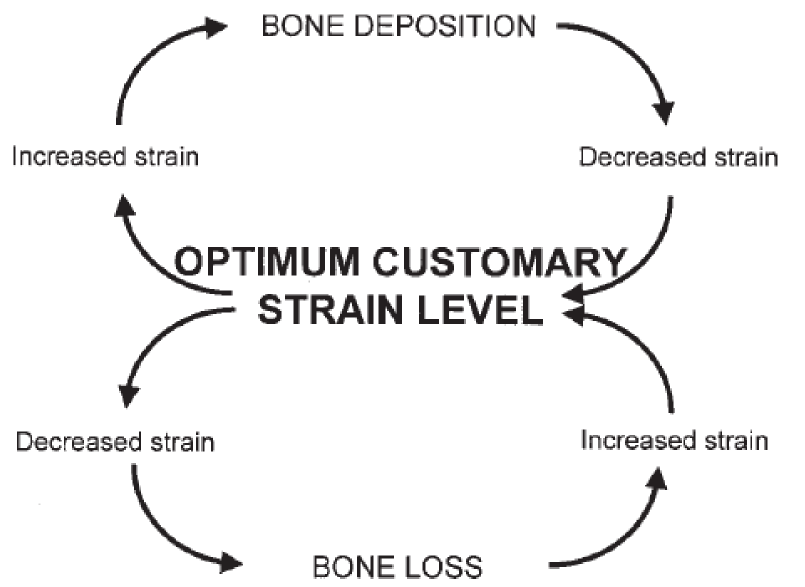
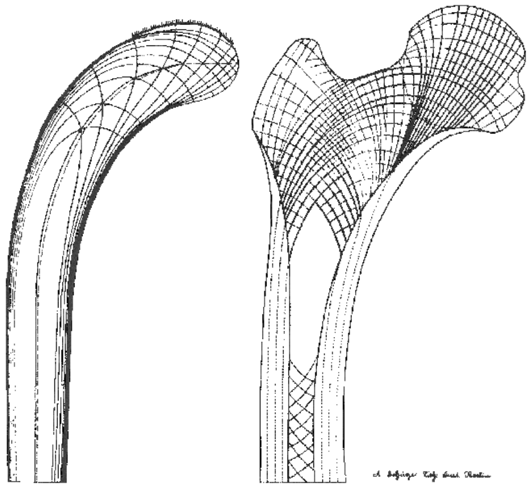
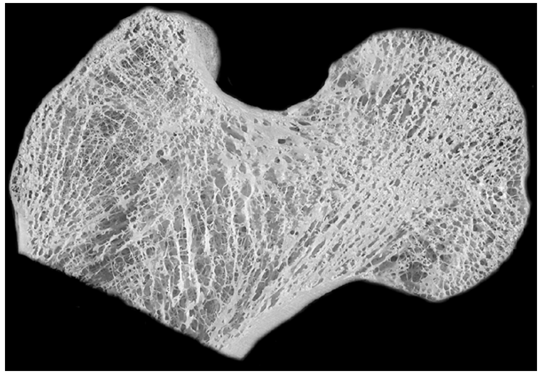
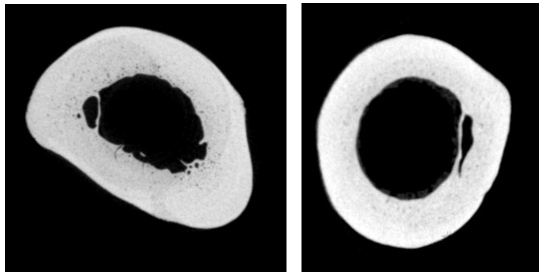


Fig. 1. Simple feedback model of bone function adaptation (from Lanyon, 1982).



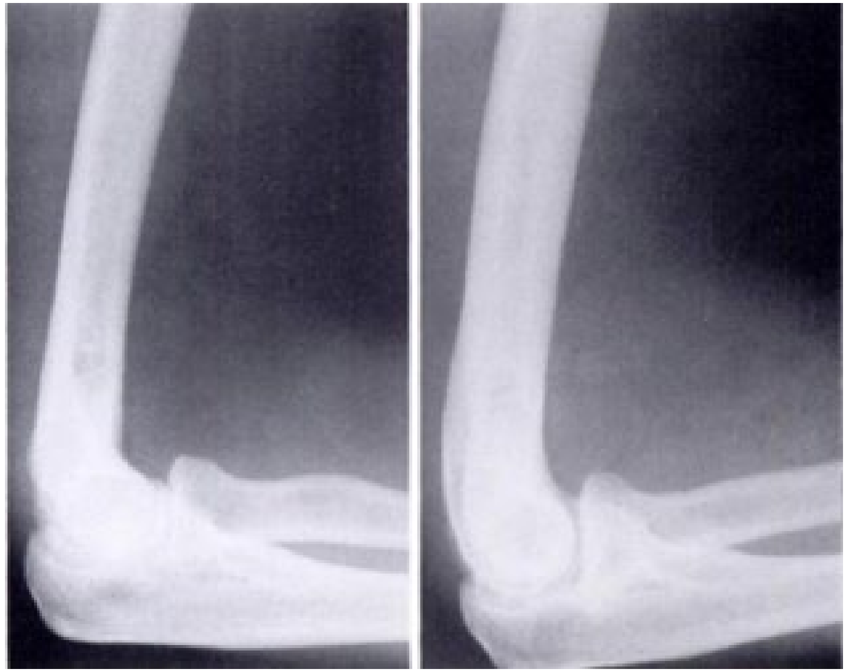




What Does Tennis Have To Do With Anthropology?

- Classic study of tennis player's arm bones
- Bone in the playing arm was more robust than non-playing arm
- Repetitive asymmetrical use leads to bone accrual

Non-playing arm Playing arm



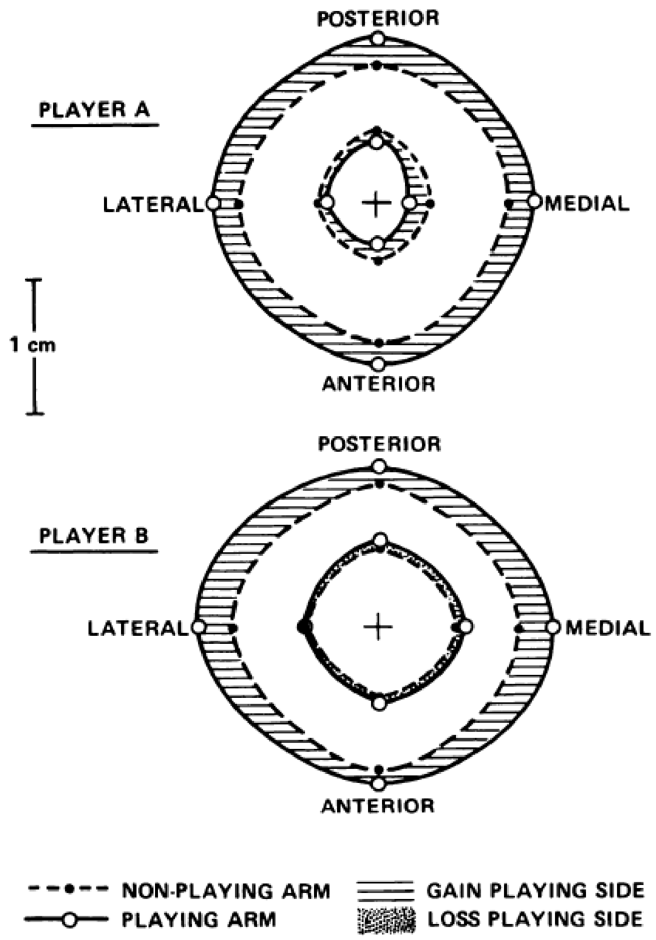


FIG. 4

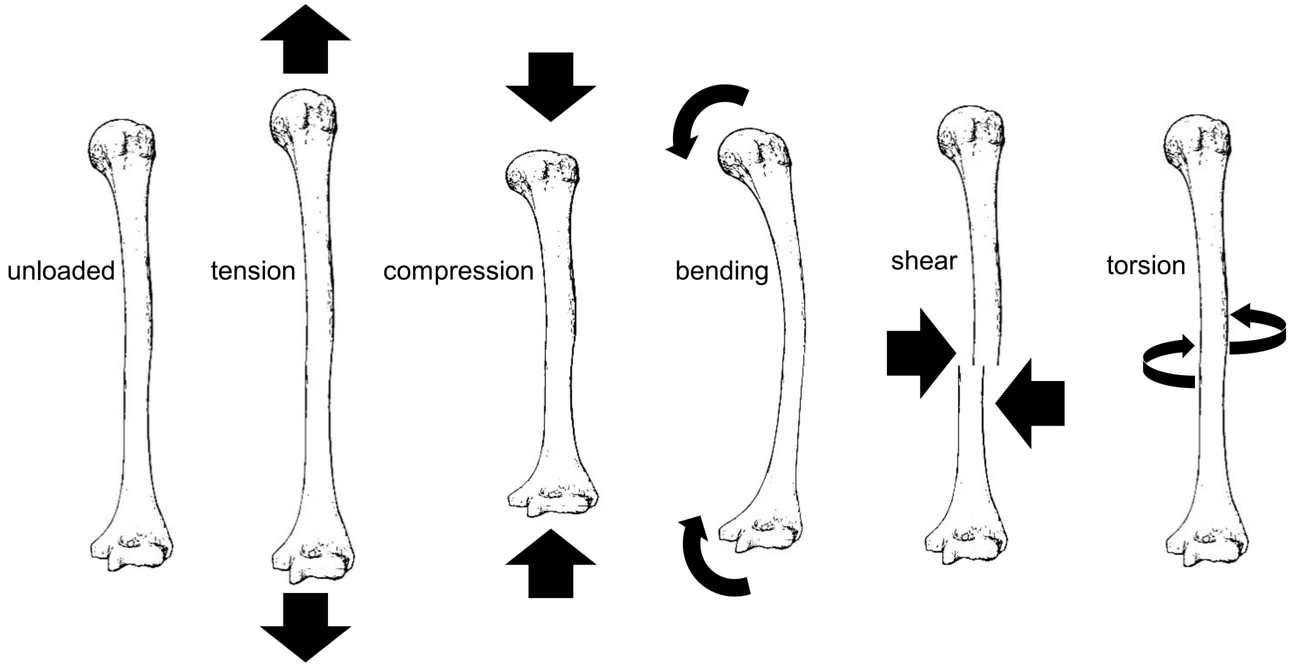
Jones et al., 1977

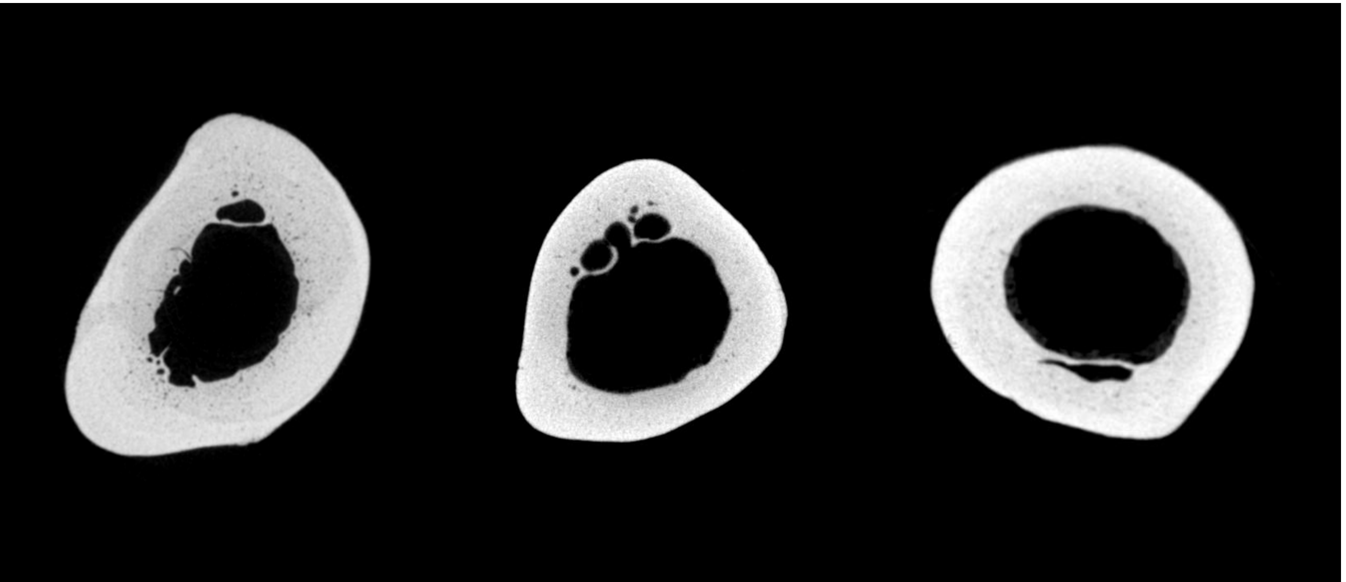
Thought Experiment

- How do different structures respond to bending loads?
- What makes some shapes better than others at resisting bending?

Engineering Beam Theory

How can we model bones as beams?





Skeletal Loading

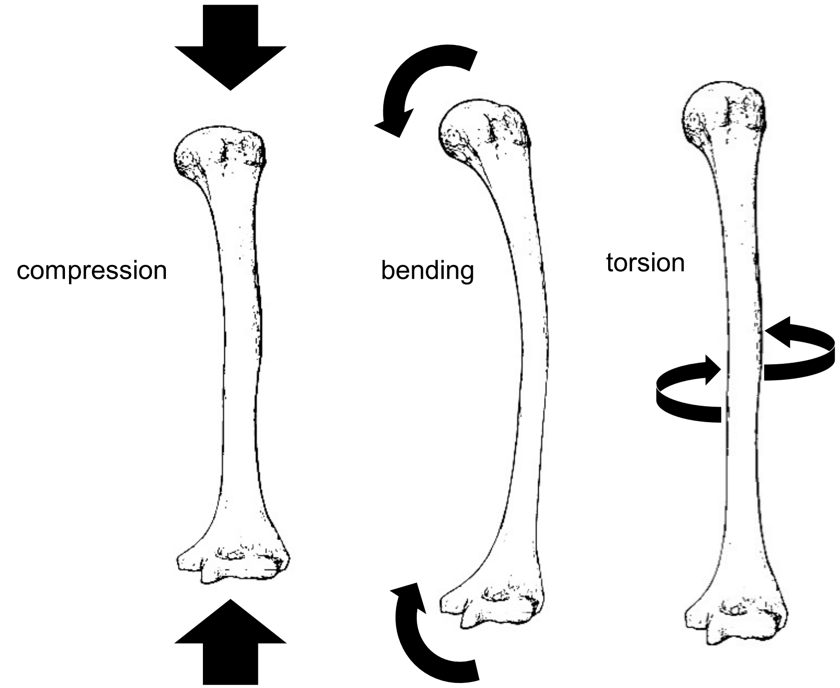
- Bones are loaded in several ways
  - Compression
  - Tension
  - Bending
  - Torsion
  - Shear
- Bone responds through growth to become stronger
- Amount and distribution of bone determines its strength

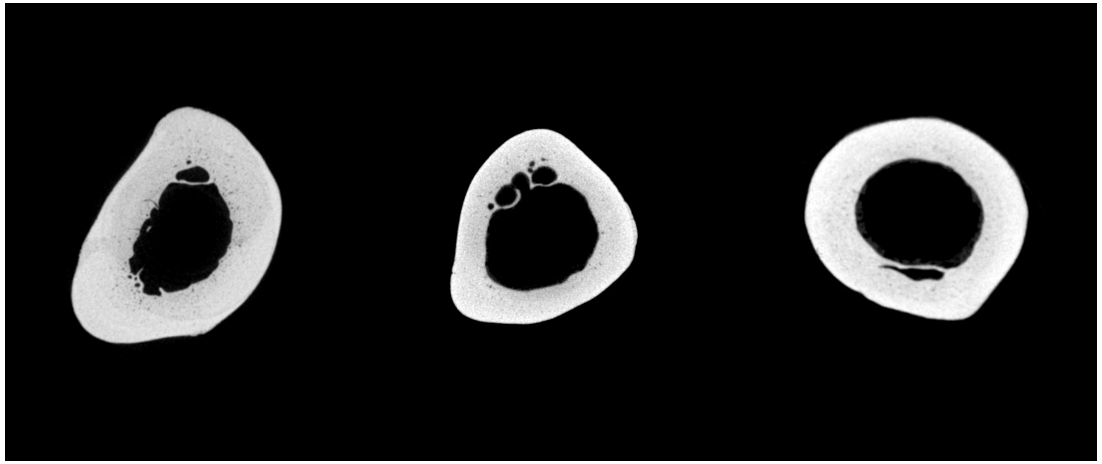
What do cross-sectional properties mean?

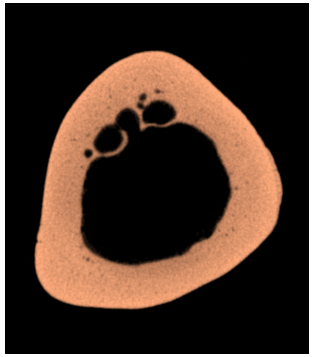
- **Cortical Bone Area** → compressive strength

- **Second moment of area** → bending strength
  - Also called moment of inertia
  - Usually measured on maximum and minimum axes
  - Measures the ability to resist bending
  - Can be used to describe shape of the cross-section

- **Polar moment of inertia** → bending and torsional strength
  - Takes into account distribution of bone around the cross-section
  - Useful for describing overall strength of bone







Cortical Bone Area (CA)

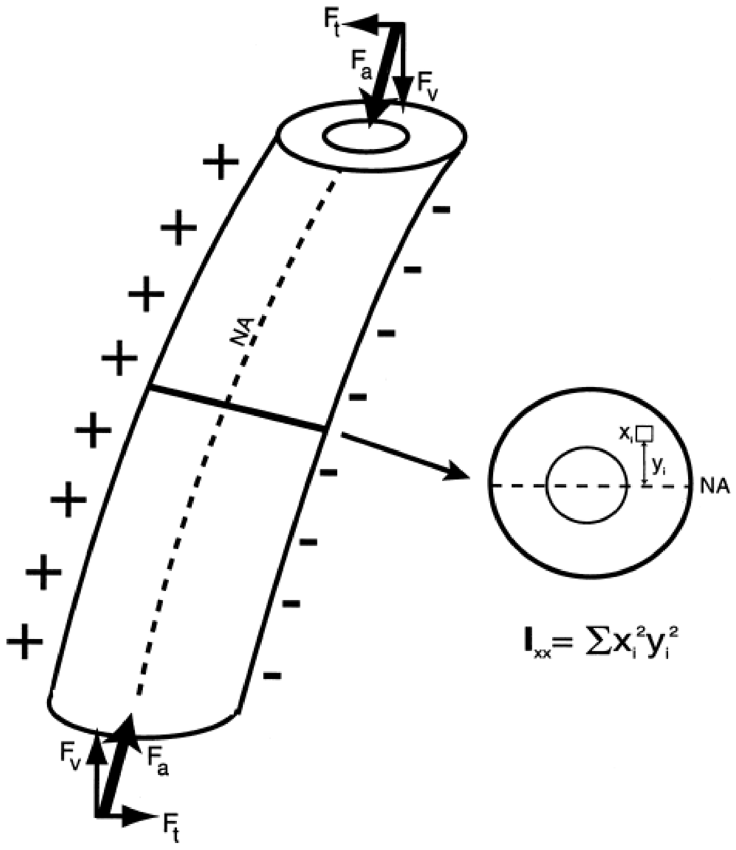
- Total area of bone in the cross-section

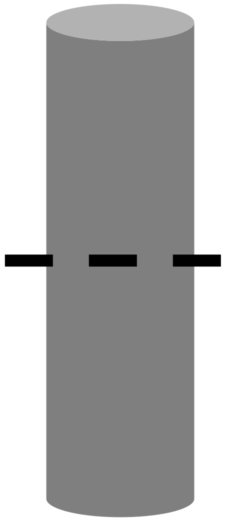
Second moment of area (I)

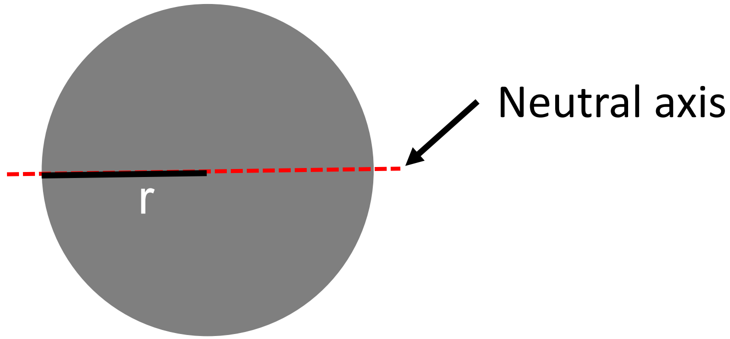
- Maximum (I_{max}) and minimum (I_{min}) axes
- I_{max}/I_{min} ratio describes shape of cross-section

Polar moment of inertia (J)

- Calculated similar to I.



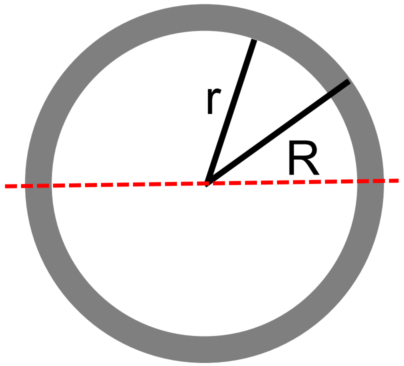




$r = 1$

$\text{Area} = \pi r^2 = 3.142$

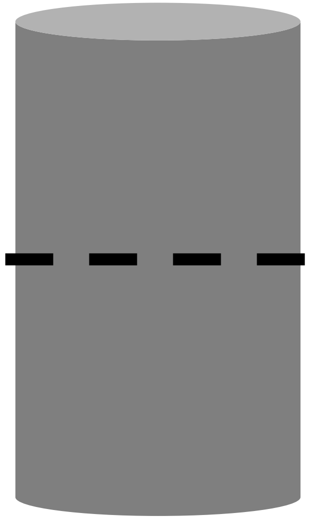
$I_{xx} = I_{yy} = \frac{\pi r^4}{4} = 0.785$

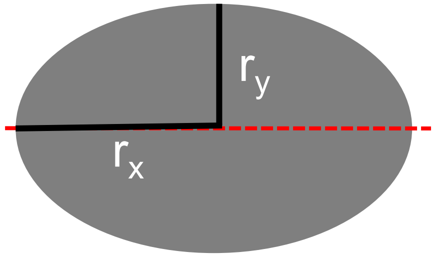


$R = 2;\ r = 1.73$

$\text{Area} = \pi(R^2 - r^2) = 3.142$

$I_{xx} = I_{yy} = \frac{\pi(R^4 - r^4)}{4} = 5.53$





$r_x = 1.33;\ r_y = 0.75$

$\text{Area} = \pi r_x r_y = 3.142$

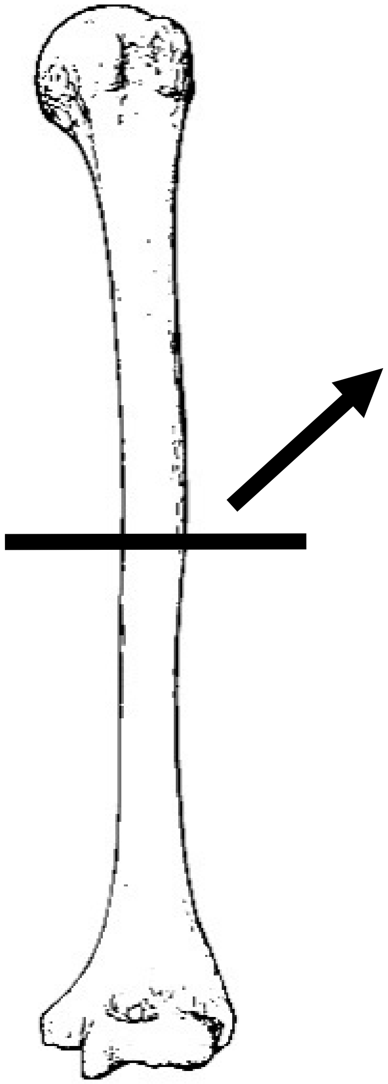
$I_{xx} = \frac{\pi r_x r_y^3}{4} = 0.442$

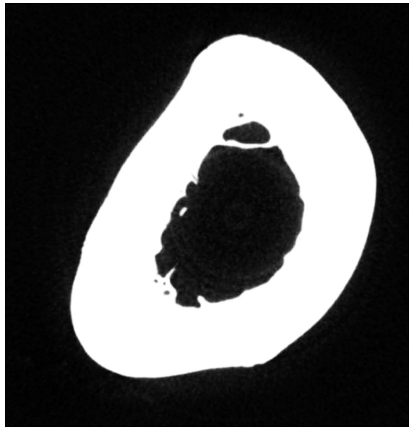
$I_{yy} = \frac{\pi r_y r_x^3}{4} = 1.396$

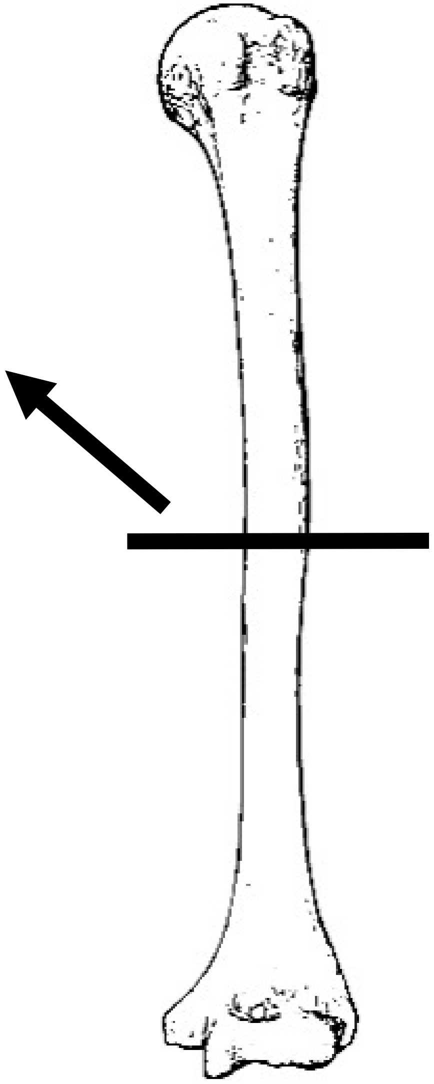
Measurements

- Area – how much material?
- Distribution – where is the material?
  - Second moment of area
  - Polar moment of area

Example

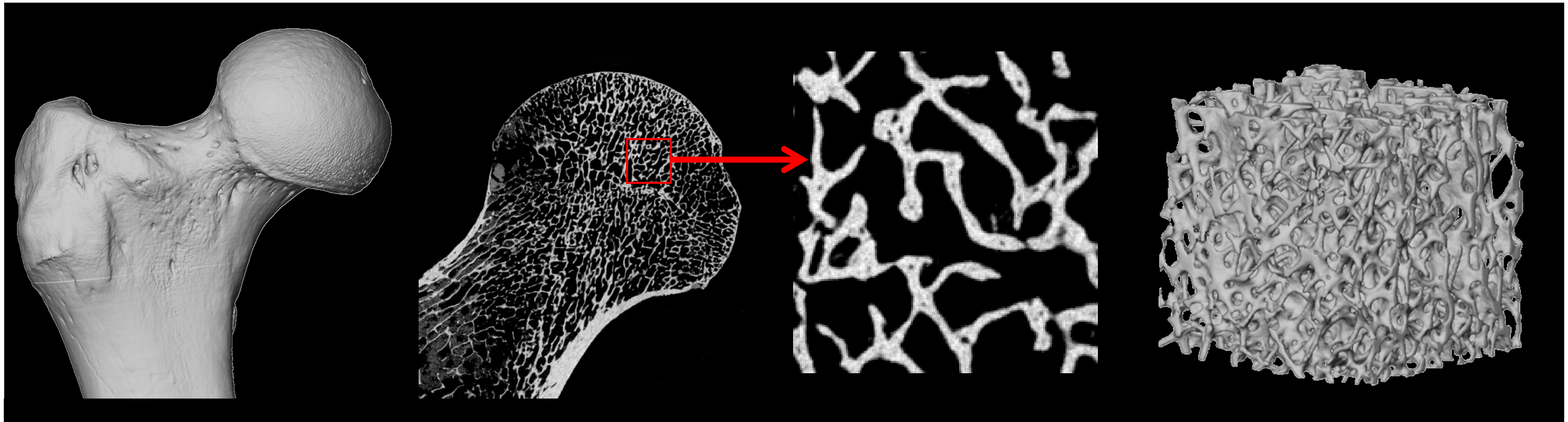






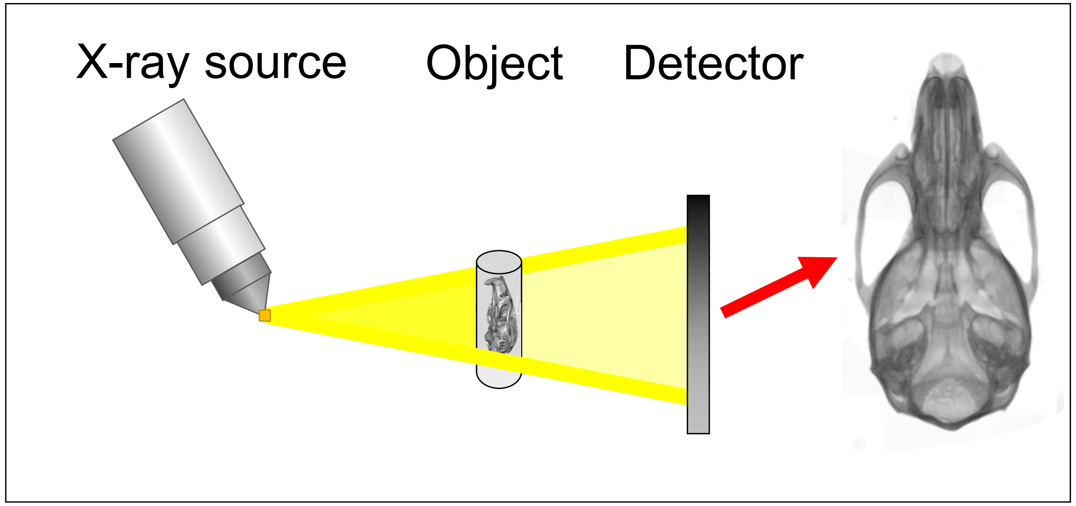
168.04	**Cortical Area**	173.40
5412.54	I_{max}	5810.00
2657.38	I_{min}	4612.26
2.04	I_{max}/I_{min}	1.26
8070.86	**Polar Moment**	10422.16

Microcomputed tomography (μCT)

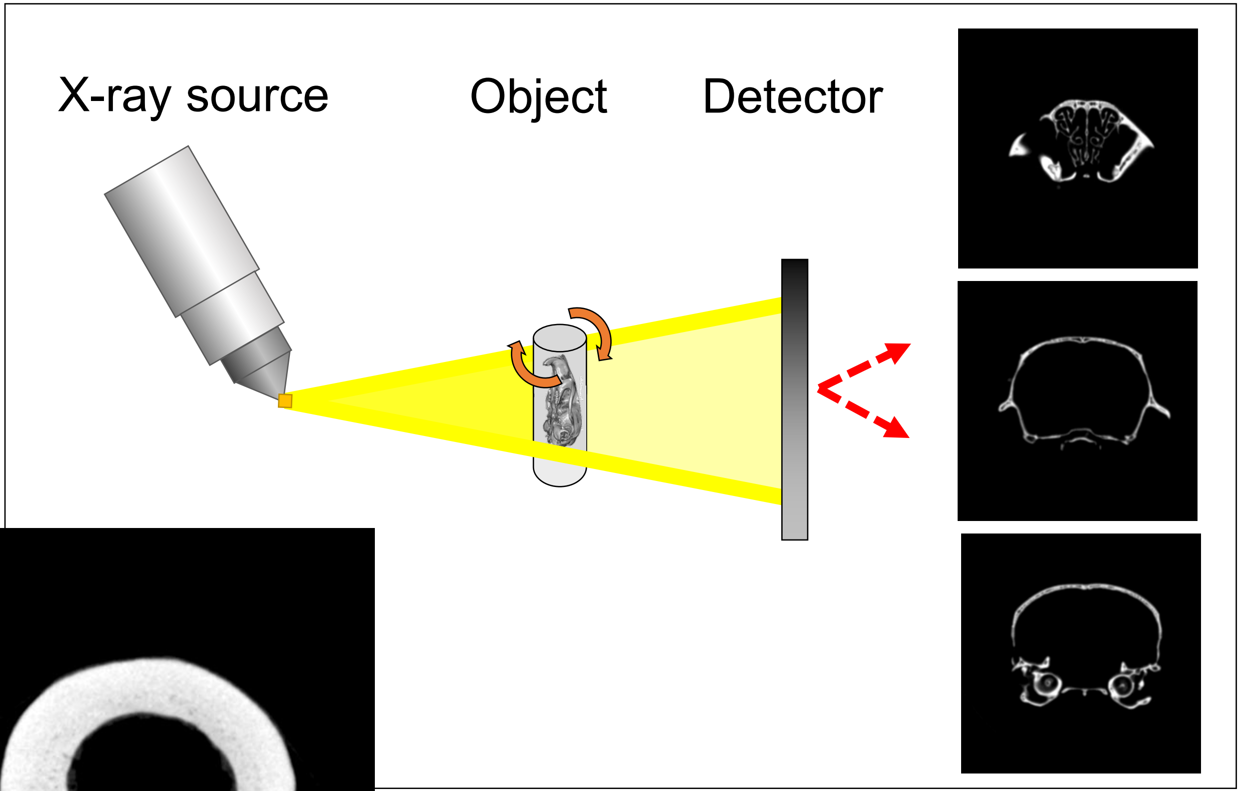


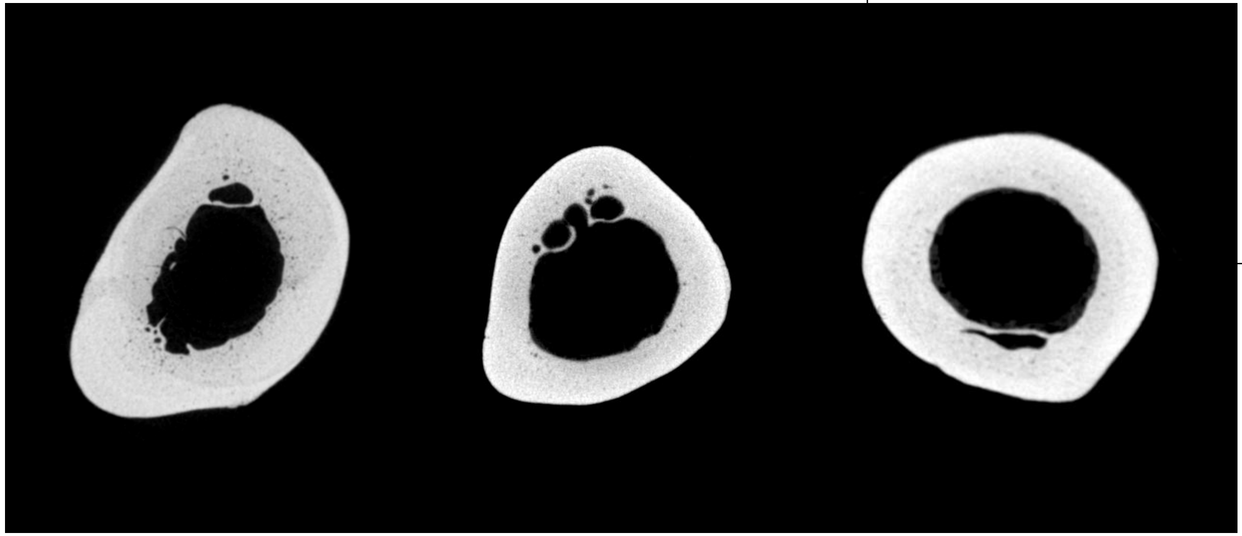
X-Ray Technologies

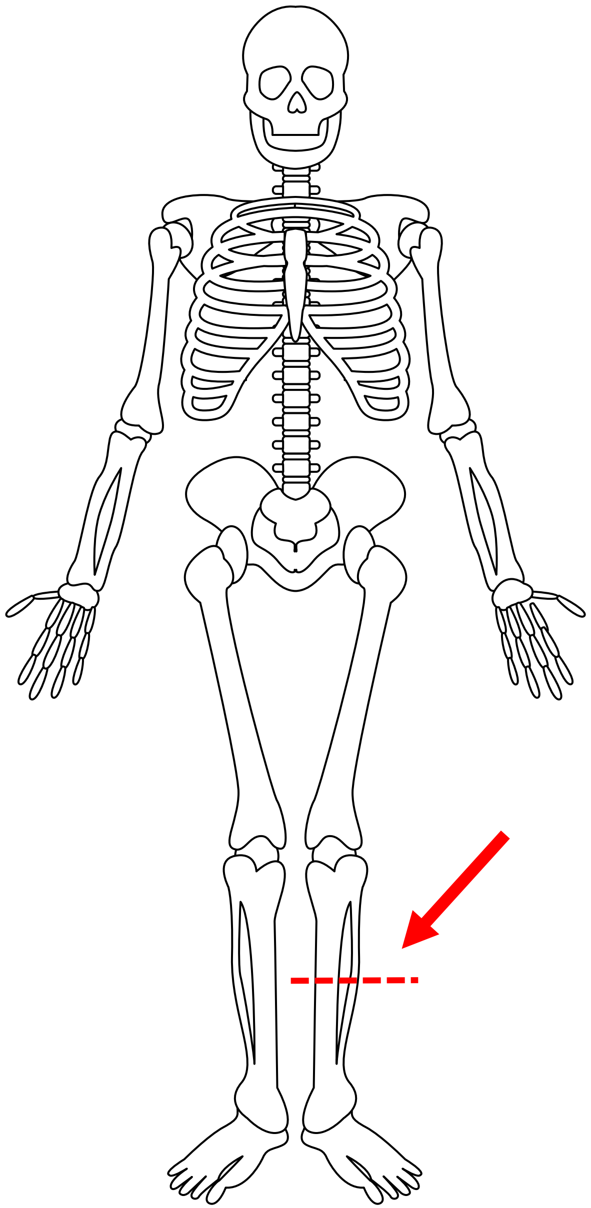
X-Ray



Computed Tomography







Evidence from athletes

- Comparison of bone strength and shape in the tibia
- Three groups
  - Field hockey
  - Long distance runners
  - Non-athlete controls
- Predictions?

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Intensity, Repetitiveness, and Directionality of Habitual Adolescent Mobility Patterns Influence the Tibial Diaphysis Morphology of Athletes

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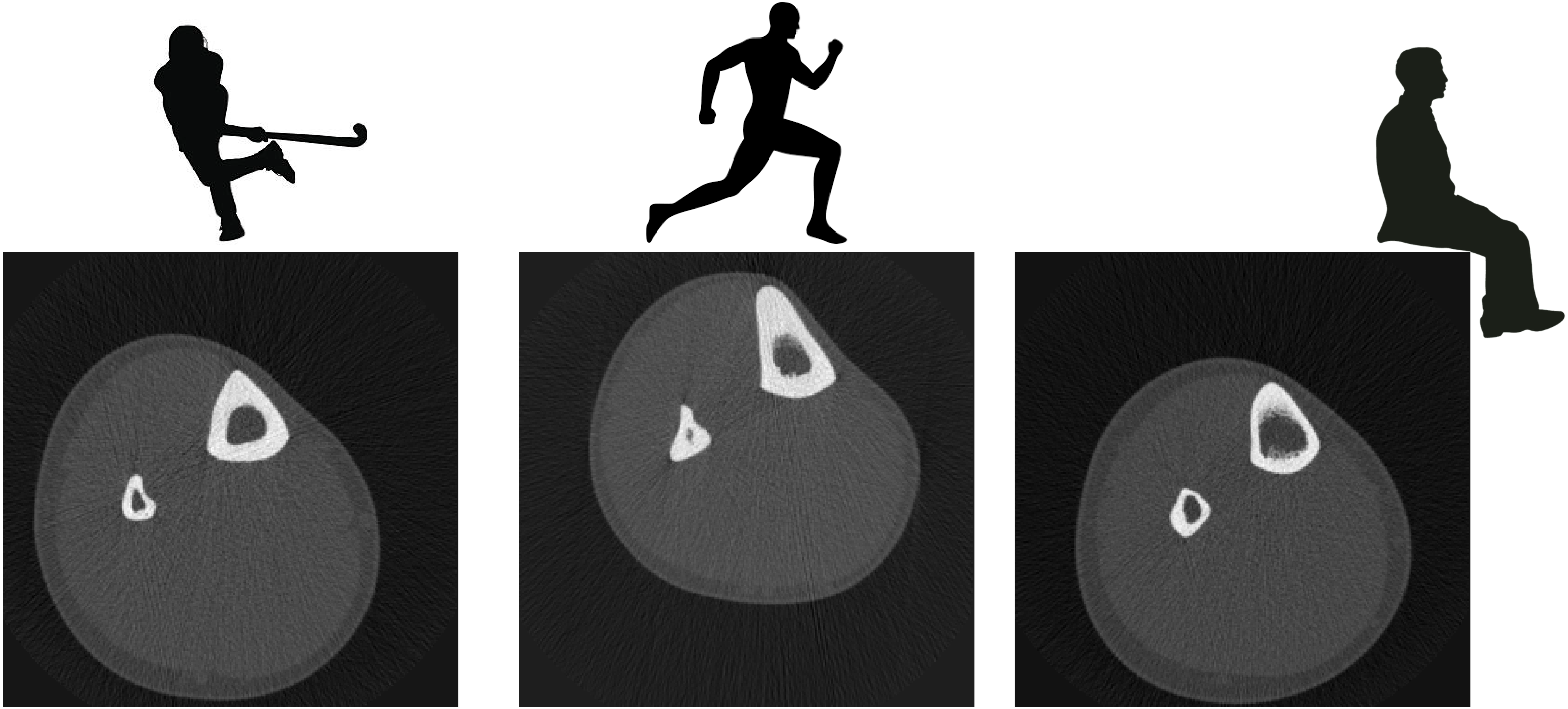
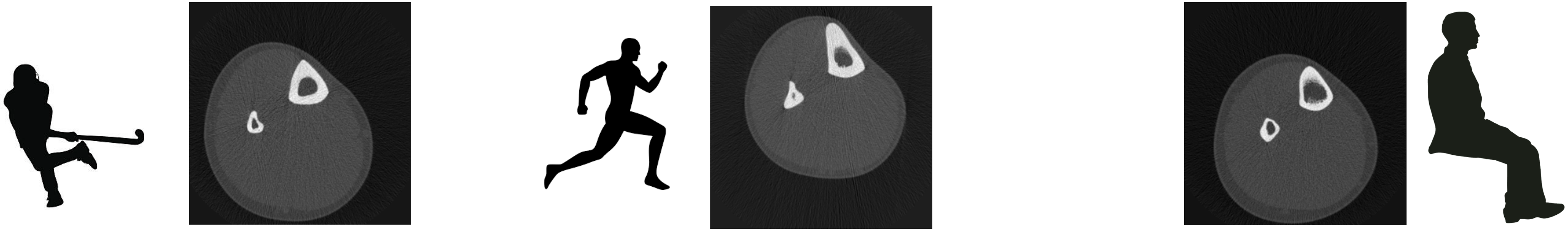
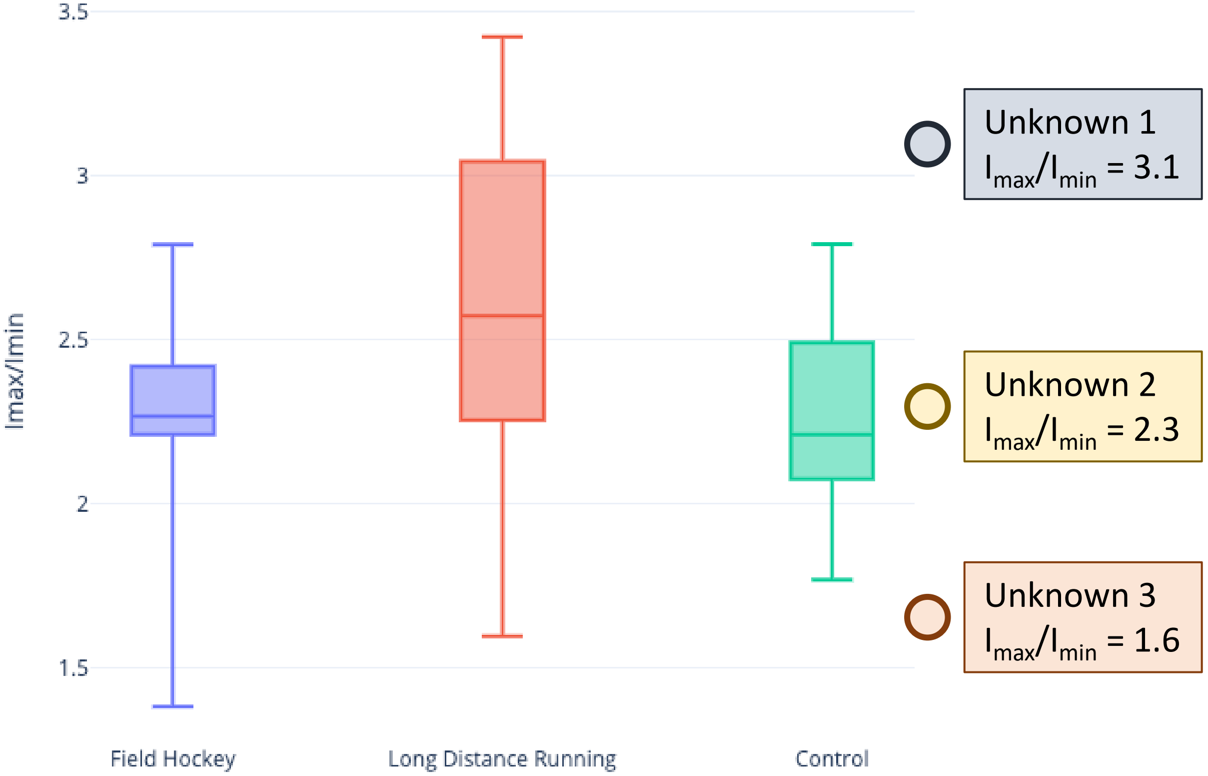


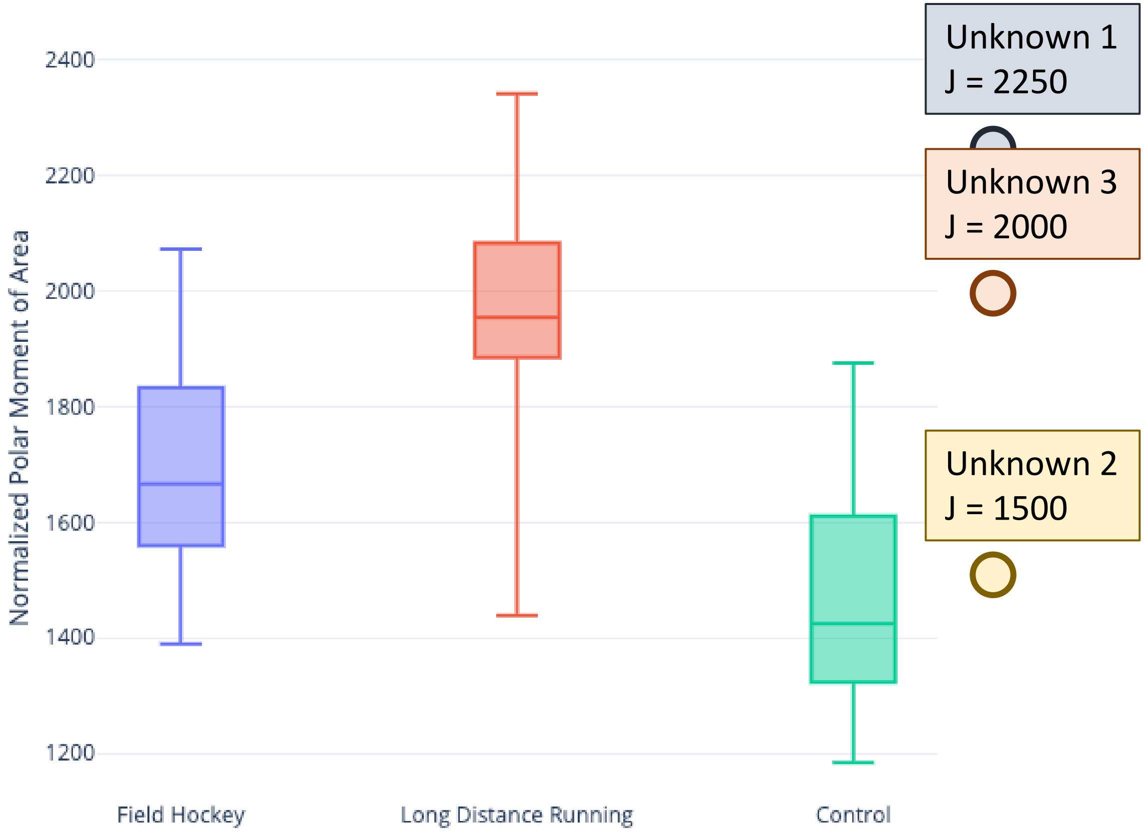
Field hockey　Long distance runners　Non-athlete controls



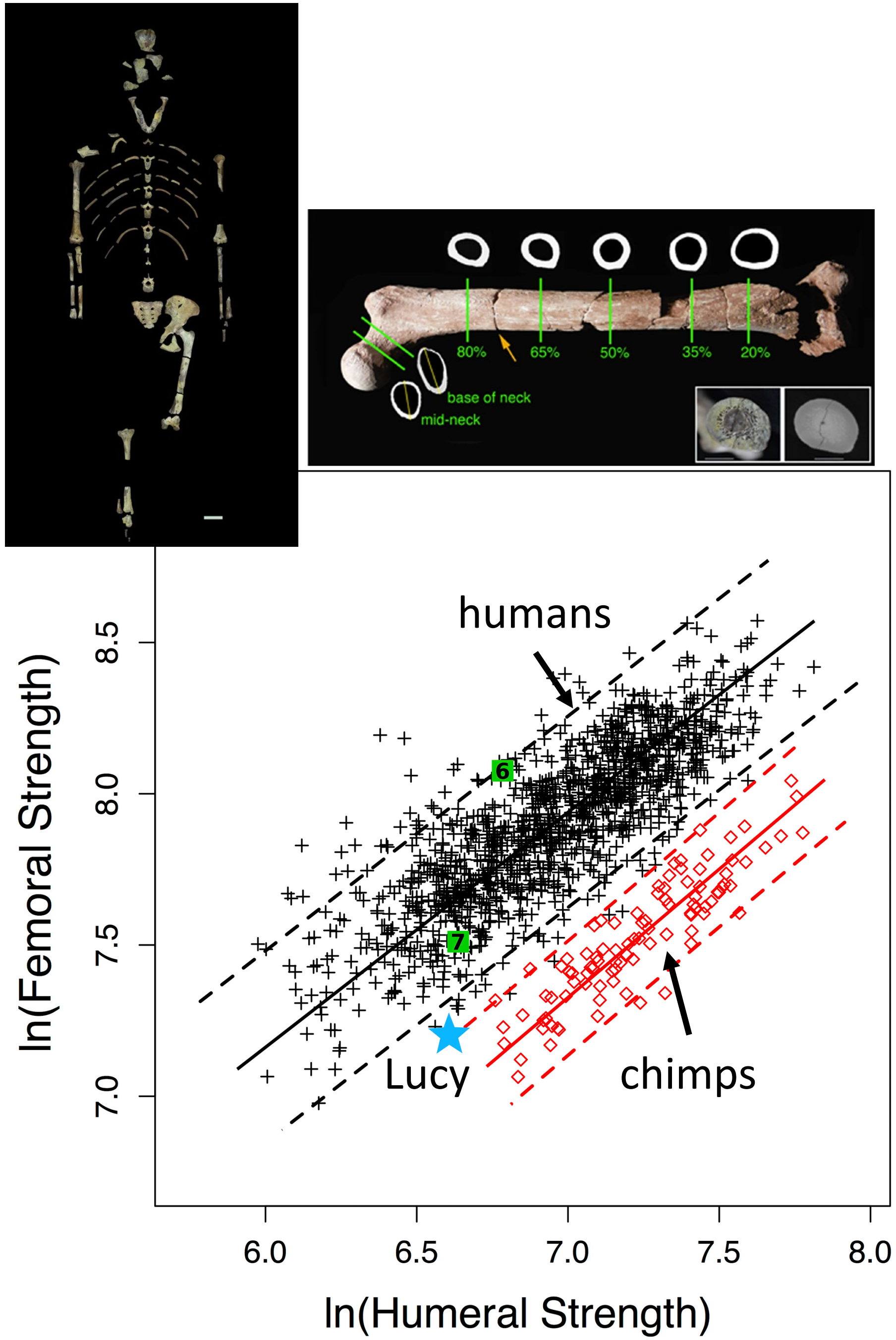
Tibia Shape (I_{max}/I_{min})



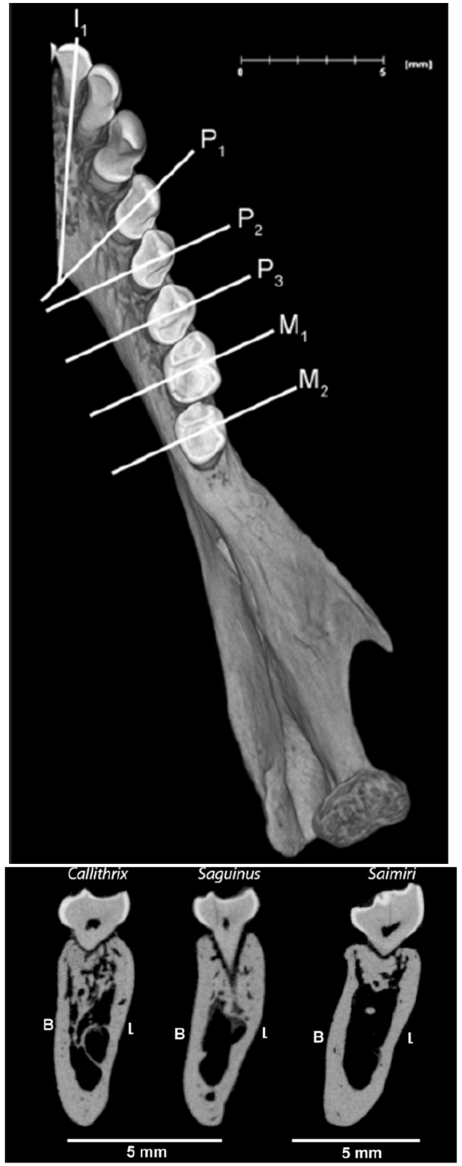
Tibia Robusticity (Polar Moment)

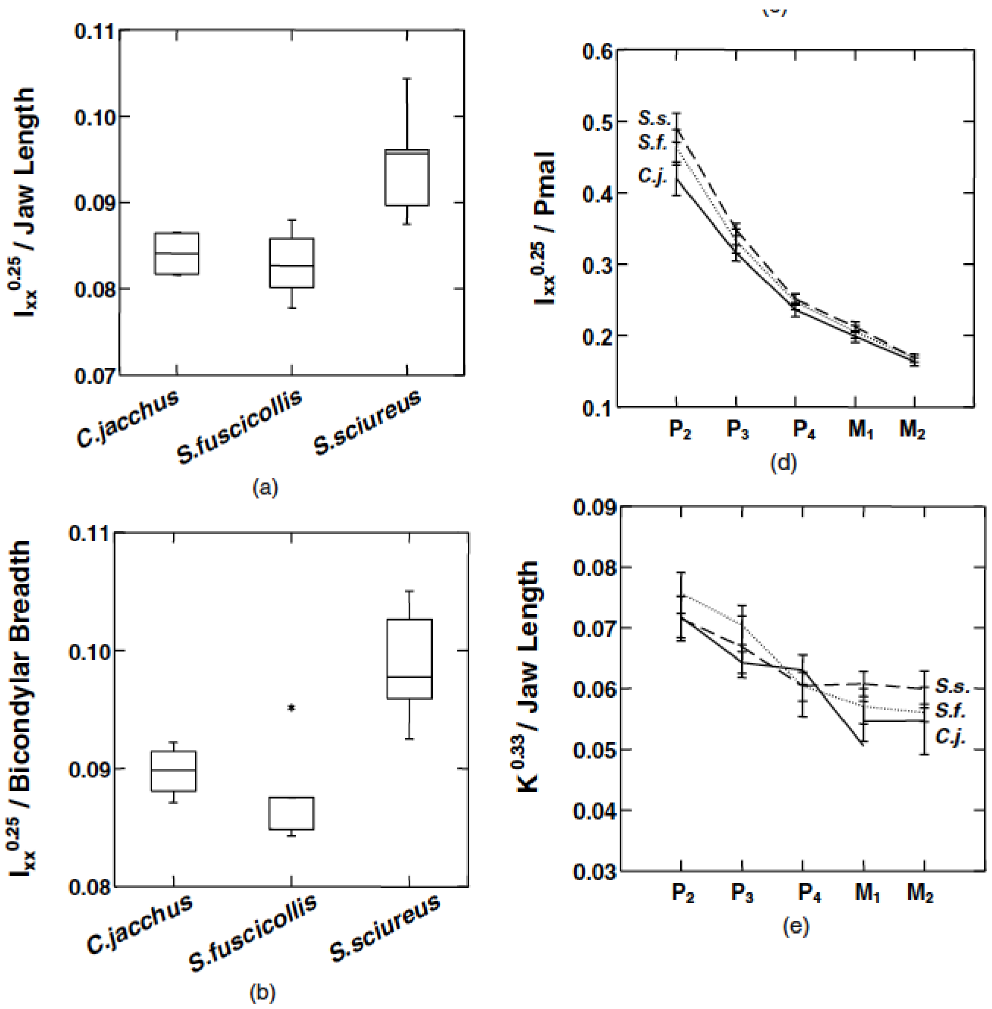


Between Species Examples



Ruff et al., 2016

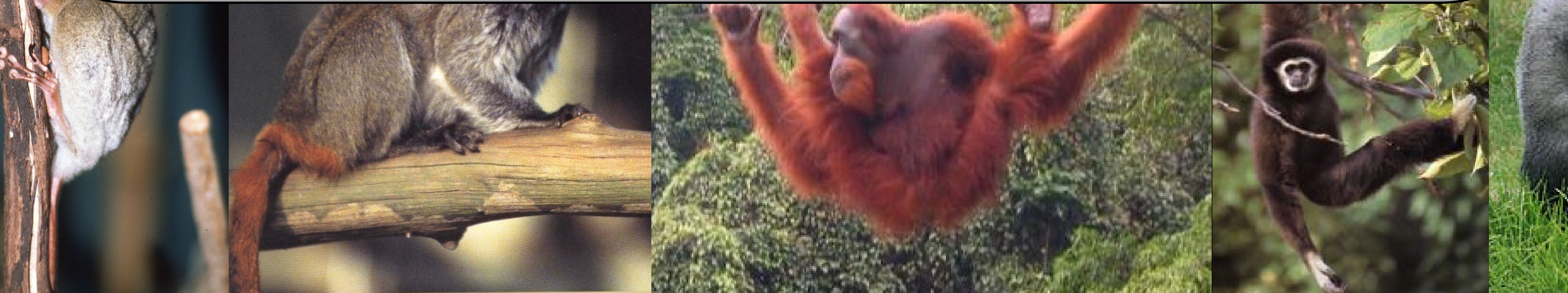




Vinyard and Ryan 2006

Questions

- How can we use the concept of bone functional adaptation and engineering beam theory to understand primate locomotion?
- How would you design a study to interpret the behavior of an unknown fossil specimen or a past human population?
- How is bone functional adaptation and skeletal plasticity potentially relevant to human bone health?



Next Steps

- Tutorial on how to collect cross-sectional geometry data
- Activity:
  - Collect cross-sectional properties for your mystery specimens
  - Plot and compare your results with data from other primates
  - Use those plots to interpret possible behavioral patterns for your mystery specimen